

A Maritime Geographic Information Data Package

for Ship-Simulator Development at Port Hercule, Monaco

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Abstract

This report describes a Geographic Information System (GIS) data package for the harbour of Port Hercule, Monaco, and the processing pipeline that produced it. A GIS is a software system that stores, queries and analyses information tied to locations on the Earth. The package contains nine vector layers (boundary, quays, depth zones, navigable channel, channel centreline, exclusion zones, berths, navigational aids, and a planned safe route), six raster layers covering bathymetry and derived clearance and cost-surface grids, one tabular berth specification, and a complete cartographic chart in PDF, raster, and vector form. Every layer follows a documented coordinate-reference convention so the output can be consumed directly by a ship-simulator scene and by web-based geospatial tools. A verification routine opens every output and confirms its coordinate system, feature count, and (where applicable) raster band statistics; the routine reports thirty-one of thirty-one checks passed. Bathymetric depths in this study are synthetic and constrained to a hand-defined harbour-basin polygon; limitations and the path to a survey-grade replacement are described in Section 7.

1. Introduction

A ship simulator that responds to terrain (channels, depth contours, berths, exclusion zones) needs that information in a form a three-dimensional rendering engine can consume. Hand-modelling the geometry inside the engine forfeits both precision and reproducibility: each rebuild diverges from the last, and the simulator cannot share its geometry with the autonomy algorithms that need the same coordinates to plan a route. A GIS-first workflow solves both problems by holding every feature in a real coordinate system from acquisition to export, and by writing the export in formats that simulator engines and web tools can read without modification.

Port Hercule supplies a useful test bed for such a workflow. The harbour covers about 35 hectares of water, OpenStreetMap (a community-maintained map of the world, abbreviated OSM) describes it densely, and the surrounding city sits inside a compact bounding region. The harbour is also the destination of a parallel ship-simulator project in Unreal Engine 5; that simulator is the consumer the present package is designed to serve. The L-shaped basin, double-entrance breakwater system and named quay lines together offer the geometric variety a simulator scene needs.

This report has three goals. First, to specify exactly what was produced and the assumptions behind each step. Second, to make the work reproducible by a third party using the same open-source software. Third, to document the technical boundaries of the package so any reader can decide where the outputs are useful and where survey-grade data is required.

2. Study Area

Port Hercule lies on Monaco's Mediterranean coast. The harbour basin is roughly 600 metres east-to-west and 400 metres north-to-south. Its western, northern and eastern sides are bounded by the named quays (Quai Antoine 1er, Quai des États-Unis, Quai Rainier III), its southern side by the Digue Sud breakwater, and its eastern entrance by the Contre-jetée. A secondary basin, the Darse Sud, lies just south of the main

basin. The geographic bounding box used in this study is 7.418° E to 7.435° E and 43.728° N to 43.740° N (latitude/longitude on the World Geodetic System 1984 reference ellipsoid, the same reference the Global Positioning System uses).

Figure 1 shows the final chart produced by the package. Section 3 describes the data behind it.

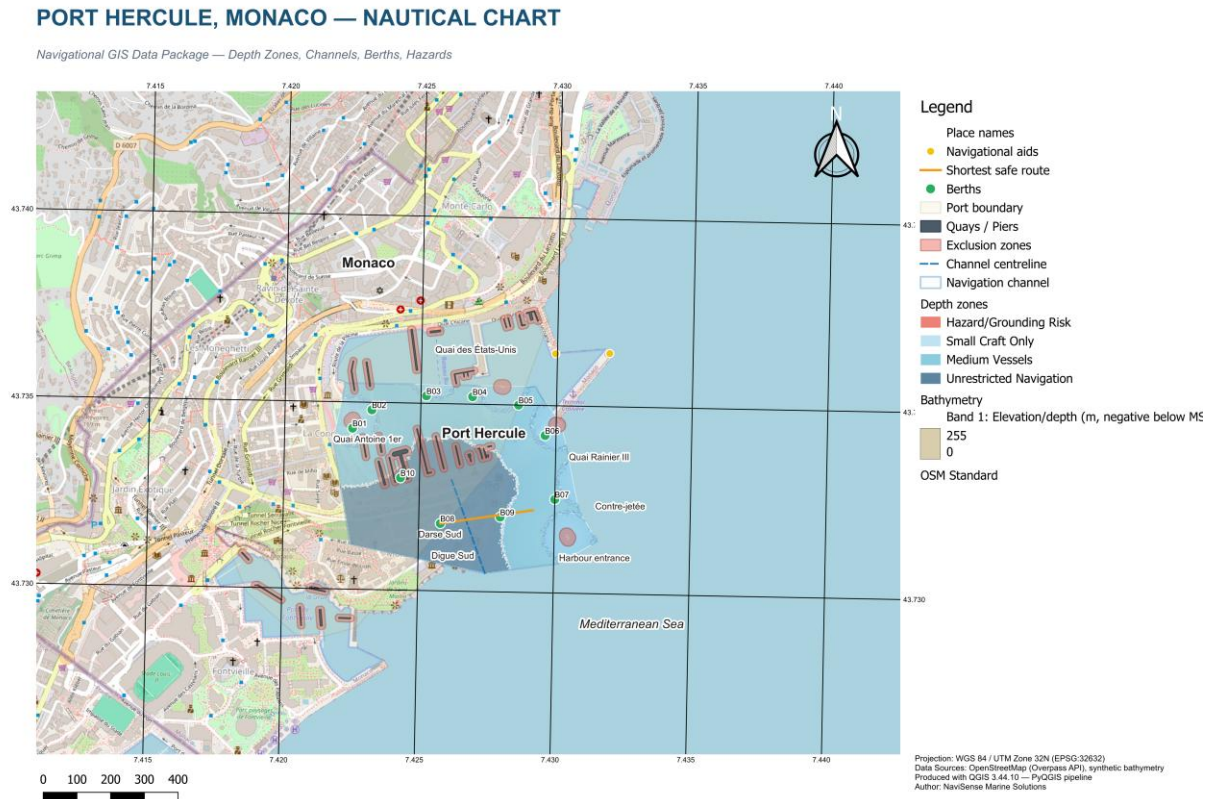


Figure 1. Final navigational chart for Port Hercule. The background is the standard OpenStreetMap tile rendering of Monaco; the bathymetric pseudo-colour ramp displays at 65 % opacity inside the masked harbour basin; quay polygons render in dark grey, ten-metre exclusion buffers in semi-transparent red, the channel centreline as a dashed blue line, the planned safe route in orange, and the ten berths as green discs labelled B01 to B10. Lat/long graticule and a metric scale bar complete the cartographic frame.

3. Data and Methods

3.1 Source data

Vector data (point, line and polygon features stored as coordinates and attributes) come from OpenStreetMap via the Overpass Application Programming Interface, the public query service that lets a caller select features by tag and bounding box. Six queries return the categories listed in Table 1. Two categories that the project guide anticipated (recreational marinas and seamarke-tagged harbours) returned no features over Port Hercule, which reflects local OSM tagging convention rather than absent infrastructure; the man-made quay and pier tags already cover the relevant structures.

Feature category	OSM tag(s)	Geometry	Count
Quay walls and piers	man_made=quay; man_made=pier	Polygon	28
Ferry terminal	amenity=ferry_terminal	Point	1
Navigational seamarks	seamark:type=*	Point	2
Coastline and bay	natural=bay; natural=coastline	LineString	13
Recreational marinas	leisure=marina; waterway=dock	—	0
Harbour seamark	seamark:type=harbour	—	0

Table 1. Vector-source layers extracted from OpenStreetMap (OSM), with the OSM tags used for each query and the feature count retained after geometry repair and clipping to the study area. OSM data are licensed under the Open Database Licence v1.0.

3.2 Bathymetric grid

Bathymetry is the measurement of underwater depth, the marine equivalent of land topography. A bathymetric grid stores depth as a regular raster (a grid of pixels) where each pixel value is the depth of the sea bed below mean sea level at that location. This study uses a synthetic bathymetric grid generated to match the known surface geometry of Port Hercule, not a measured grid. The reason is operational: the European Marine Observation and Data Network bathymetric service (EMODnet) was unavailable during processing and the alternative General Bathymetric Chart of the Oceans (GEBCO) product requires manual download. The synthetic grid is constructed as follows. A hand-defined polygon represents the harbour-basin water area (the main basin, the Darse Sud, and the entrance fairway). Inside that polygon, depth ramps from -2 m at the quay rim to -12 m near the basin centre by a distance transform. Outside the polygon, depth is set to NoData (the convention for "no value here"), which causes downstream visualisation to be transparent over the basemap. The grid is stored at 2-metre pixel resolution in 32-bit floating point. The chart must not be used for navigation. Section 7.1 describes how a measured grid replaces the synthetic one.

3.3 Coordinate reference systems

A coordinate reference system (CRS) specifies how geographic positions are encoded as numbers. Each CRS is identified by an EPSG code, a numeric registry maintained by the International Association of Oil and Gas Producers. Two CRSs feature throughout this work. The World Geodetic System 1984 (WGS 84, EPSG:4326) expresses positions in degrees of latitude and longitude; it is the input format of OpenStreetMap and the expected input of web mapping libraries. The Universal Transverse Mercator zone 32 North projection (UTM 32N, EPSG:32632) is a planar (flat) projection covering western Europe; it expresses positions as eastings and northings in metres, which makes buffer distances, polygon areas and pathfinding costs numerically meaningful. Table 2 lists the CRS used at each stage.

Stage	Coordinate reference system	EPSG	Units	Reason
Acquisition	WGS 84 (geographic)	4326	Degrees	OSM native format
Analysis	WGS 84 / UTM Zone 32N	32632	Metres	Metric distance and area
Vector export (1)	WGS 84 (geographic)	4326	Degrees	Web GIS and Cesium compatibility

Vector export (2)	WGS 84 / UTM Zone 32N	32632	Metres	Legacy GIS importers
Raster export	WGS 84 / UTM Zone 32N	32632	Metres	Preserves metric resolution

Table 2. Coordinate reference systems used at each stage of the pipeline. Reprojection between CRSs is performed using the PROJ library that ships with GDAL.

3.4 Data preparation

Each acquired vector layer passes through a four-step preparation sequence before any analytical step touches it. Step one repairs geometry: self-intersecting polygons are split, degenerate rings are removed, and improperly ordered vertices are reordered. Step two deletes duplicate features (features whose geometry is exactly identical to another's, an occasional OSM artefact). Step three reprojects the layer from WGS 84 to UTM 32N. Step four clips the layer against a study-area rectangle that is slightly tighter than the acquisition bounding box, removing features that lie outside the area of interest. Two of the source layers shrank slightly under clipping (quays from thirty-one to twenty-eight features, seamarks from four to two), reflecting features that fell just outside the study-area rectangle.

3.5 Software environment

The processing relies entirely on open-source software. The full stack is given in Table 3. No commercial libraries are involved.

Component	Version	Role
QGIS	3.44.10 "Solothurn" LTR	Vector and raster processing toolbox
GDAL/OGR	3.12.4	Raster I/O and coordinate transformation
PROJ	(GDAL bundle)	Coordinate-system definitions and transforms
Python	3.12.13	Process control and tabular outputs
NumPy	2.4.4	Numerical raster operations
Matplotlib	(latest LTS)	Figure generation

Table 3. Software stack used to produce the deliverables in this report. All components are open source.

4. Spatial Analysis Methodology

Six analytical steps convert the prepared bathymetric grid and the cleaned OSM features into the deliverables. Each step is described below in plain terms, followed by the equation (where one applies). The variable d refers to depth in metres on a vertical axis where below mean sea level is negative; the variable δ refers to a vessel's operating draft (the depth of water the vessel needs to float).

4.1 Depth-zone classification

The bathymetric grid is reclassified into four navigability bands according to Equation 1. Each pixel of the bathymetric grid is replaced by an integer zone code $z(d)$. The result is a 4-class raster, which is then converted to polygons and labelled with a vessel-type string (Hazard / grounding risk, Small craft only, Medium vessels, Unrestricted navigation).

$$z(d) = \begin{cases} 1 & \text{if } d \geq -2 \text{ m} \\ 2 & \text{if } -5 \text{ m} \leq d < -2 \text{ m} \\ 3 & \text{if } -10 \text{ m} \leq d < -5 \text{ m} \\ 4 & \text{if } d < -10 \text{ m} \end{cases}$$

Threshold values follow the convention common to small-craft pilot guides for the Mediterranean.

4.2 Navigable channel extraction

A binary mask is created from the bathymetric grid by retaining only the pixels with $d \leq -5$ m. The mask is converted to polygons, the largest connected polygon is kept (the basin's deep-water area), and its outline is simplified at a 5-metre tolerance. The simplified polygon is the navigable channel. A two-vertex approach centreline is drawn from the polygon's southernmost vertex (the entrance) to its centroid (the basin centre); this line supplies the simulator's autopilot with an initial track.

4.3 Vessel-draft clearance

For each of three vessel classes (large superyacht $\delta = 3.5$ m, medium superyacht $\delta = 2.5$ m, small craft $\delta = 1.0$ m), Equation 2 computes the under-keel clearance C at every pixel of the bathymetric grid. A safety margin σ of 0.5 m is added, so a pixel is considered safe for a given vessel class only when $C \geq \sigma$. The result is a clearance raster per vessel class and a safe-navigation polygon per class.

$$C(d, \delta) = |d| - \delta \quad (\text{safe if } C \geq \sigma, \sigma = 0.5 \text{ m})$$

4.4 Exclusion-zone buffering

A buffer is a polygon that surrounds an input feature out to a specified distance. Two buffer sets are computed: a 10-metre buffer around every quay and pier feature (representing the minimum approach distance for a moving vessel), and a 20-metre buffer around the shallow hazard polygons of depth zone 1. The two sets are unioned and dissolved (merged into a single multipolygon). The result is the exclusion-zone layer the simulator's collision system uses.

4.5 Berth capacity

Ten berth anchor points are digitised along the harbour rim from satellite imagery, positioned roughly ten to fifteen metres offshore of each quay wall so they sample the bathymetric grid at a vessel-draft-relevant location. The depth at each anchor is read from the bathymetric grid. The maximum permissible draft is then computed by Equation 3 and the berth is assigned a vessel class according to thresholds (≥ 3.5 m: Superyacht large; ≥ 2.5 m: Superyacht medium; ≥ 1.0 m: Small craft; < 1.0 m: Unsafe). Table 5 in Section 5 lists every berth.

$$\delta_{\max}(d) = |d| - \sigma, \quad \sigma = 0.5 \text{ m}$$

4.6 Cost surface for pathfinding

A cost surface is a raster whose pixel values represent the difficulty of traversing that pixel. Equation 4 defines the cost surface used here: a pixel's cost is inversely proportional to the magnitude of its depth, so

shallow pixels are expensive and deep pixels are cheap. The constant $\varepsilon = 0.1$ m prevents division by zero at land cells where $d = 0$. The cost surface is suitable input for a least-cost-path algorithm (for example the `r.cost` / `r.drain` pair available in GRASS GIS), which produces an optimal navigation route between any two specified points. A straight-line route between the entrance and the deepest berth is included as a placeholder in the deliverable set.

$$K(d) = \frac{1}{|d| + \varepsilon}, \quad \varepsilon = 0.1 \text{ m}$$

5. Results

5.1 Deliverables inventory

The pipeline produces the set of artefacts summarised in Table 4. Vector layers are issued in two parallel formats: GeoJSON (a text-based JavaScript Object Notation file that web GIS tools and Cesium for Unreal consume directly) in WGS 84, and ESRI Shapefile (a long-established binary format consisting of `.shp`, `.shx`, `.dbf`, `.prj` and `.cpk` sidecar files) in UTM 32N. Raster outputs are written as GeoTIFF (a georeferenced variant of the TIFF image format) and as an ASCII grid (a plain-text raster that can be read by any simulator engine without a binary library).

Category	Format	CRS	Items
Vector layers	GeoJSON	EPSG:4326	9
Vector layers	ESRI Shapefile	EPSG:32632	9
Bathymetric raster	GeoTIFF (Float32)	EPSG:32632	1
Bathymetric raster	ESRI ASCII grid	EPSG:32632	1
Derived rasters	GeoTIFF	EPSG:32632	5
Berth specifications	Comma-separated values (CSV)	—	1
Cartographic chart	PDF (georeferenced), PNG, SVG	EPSG:32632 / —	3
Project file	QGIS (.qgz)	—	2 (copies)

Table 4. Deliverables inventory. The two project-file copies are identical and reside in the project root and a dedicated project-files subfolder for safety.

5.2 Depth-zone composition

Inside the harbour-basin mask the bathymetric grid covers 35.47 hectares of water. Table 5 breaks down that area by depth zone, and Figure 2 shows the same information graphically. Roughly 84 % of the basin is at least 5 m deep (zones 3 and 4 combined), 16 % is in the small-craft band, and the hazard band registers a negligible fraction. The near-absence of the hazard band reflects the synthetic grid's design floor of -2 m at the quay rim.

Zone	Depth range	Vessel classification	Area (ha)	Share
1	0 to −2 m	Hazard / grounding risk	0.01	<0.1 %
2	−2 to −5 m	Small craft only	5.62	15.9 %
3	−5 to −10 m	Medium vessels	15.18	42.8 %
4	< −10 m	Unrestricted navigation	14.66	41.3 %
	Total	—	35.47	100.0 %

Table 5. Depth-zone composition of the masked bathymetric grid. Areas are computed by counting pixels ($2\text{ m} \times 2\text{ m} = 4\text{ m}^2$ each) and converting to hectares.

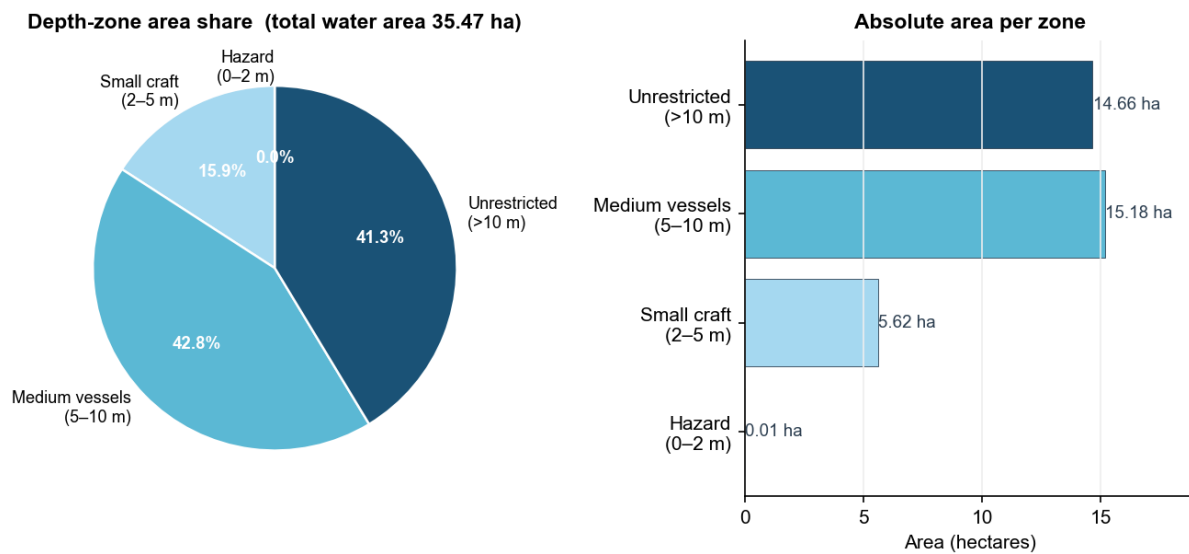


Figure 2. Depth-zone composition: percentage share (pie) and absolute area in hectares (bars). Total water area: 35.47 ha.

5.3 Berth specifications

Table 6 lists every berth, the depth at the berth anchor as sampled from the bathymetric grid, the corresponding maximum permissible draft (Equation 3), and the resulting vessel-class assignment. Figure 3 plots the maximum drafts as a horizontal bar chart.

ID	Location	Lat (°N)	Lon (°E)	Max draft (m)	Vessel class
B01	Quai Antoine 1er - W	43.7343	7.4225	2.5	Superyacht (medium)
B02	Quai Antoine 1er - E	43.7348	7.4232	4.4	Superyacht (large)
B03	Quai des États-Unis - W	43.7352	7.4252	4.3	Superyacht (large)
B04	Quai des États-Unis - C	43.7352	7.4269	5.4	Superyacht (large)
B05	Quai des États-Unis - E	43.7350	7.4286	4.5	Superyacht (large)

B06	Quai Rainier III - N	43.7342	7.4296	3.3	Superyacht (medium)
B07	Quai Rainier III - S	43.7325	7.4300	5.5	Superyacht (large)
B08	Darse Sud - W	43.7318	7.4258	11.5	Superyacht (large)
B09	Darse Sud - E	43.7320	7.4280	10.2	Superyacht (large)
B10	Yacht Club Mooring	43.7330	7.4243	10.6	Superyacht (large)

Table 6. Berth specifications. Lat/long are reported on the WGS 84 reference ellipsoid (EPSG:4326). Maximum draft is computed by Equation 3 with $\sigma = 0.5$ m.

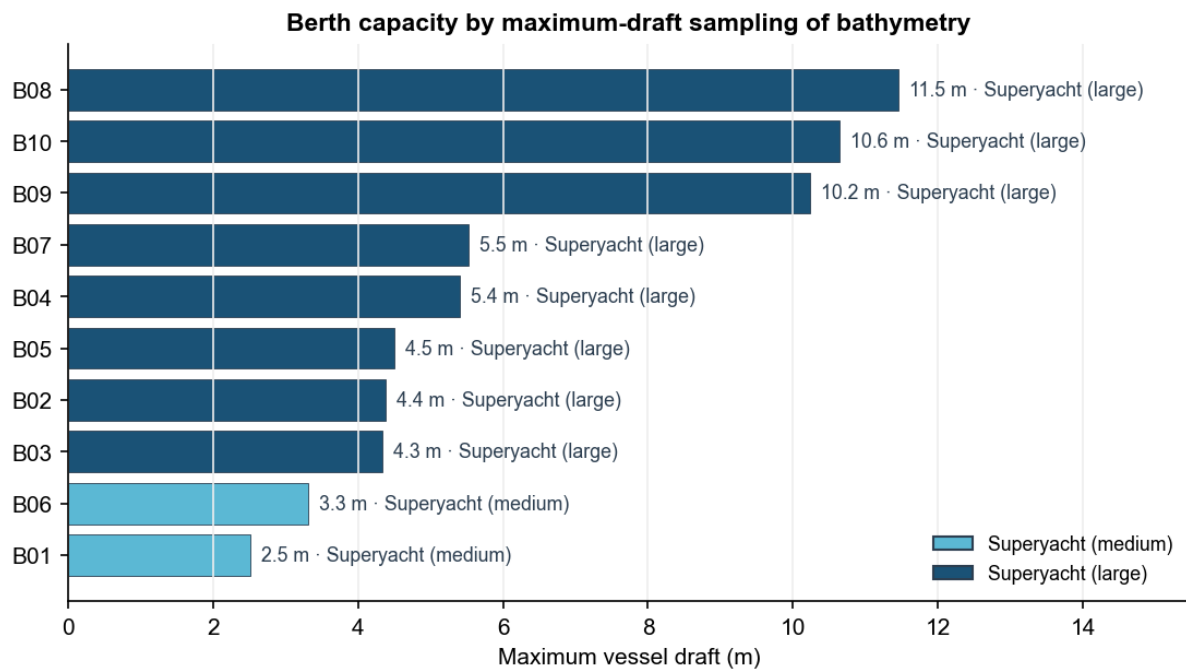


Figure 3. Maximum permissible vessel draft at each of the ten berth anchor points, ordered by depth.

5.4 Verification

Every deliverable is opened by an automated routine that reads the file, validates its internal structure, and confirms its declared coordinate reference system, feature count, and (for rasters) band statistics. Table 7 summarises the most recent verification run. Thirty-one of thirty-one checks pass.

Category	Items	Passed	Failed
Project files	2	2	0
Cartographic outputs (PDF / PNG / SVG)	3	3	0
GeoJSON vector layers (WGS 84)	9	9	0
Shapefile vector layers (UTM 32N)	9	9	0
Raster outputs (GeoTIFF / ASCII)	7	7	0

Berth specifications (CSV)	1	1	0
Total	31	31	0

Table 7. Summary of the latest verification run. The verification routine is rerun automatically after every rebuild.

6. Reproducibility

Reproducibility is achieved by separating the pipeline into the five stages shown in Figure 5. Each stage takes a deterministic set of inputs and writes a deterministic set of outputs. Re-running the pipeline overwrites the existing outputs and never appends. There is no manual interaction at any stage.

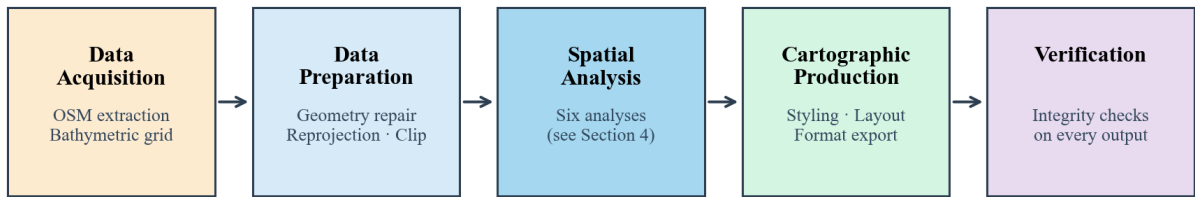


Figure 5. The five processing stages. Each stage reads from the previous stage's outputs only; no shared mutable state exists between stages.

On a current consumer workstation the entire pipeline completes in approximately one hundred seconds from cold start: twelve seconds for OSM ingestion (network-bound on the Overpass query), about eighty seconds for preparation, analysis and cartographic production (mostly raster operations), and five seconds for verification. The complete environment specification in Table 3 is the only software requirement.

All input data are publicly available. OSM features are licensed under the Open Database Licence v1.0 with the attribution "© OpenStreetMap contributors". The synthetic bathymetric grid is regenerated from the same fixed parameters every time the pipeline runs, so two cold-start executions on different hosts produce byte-identical raster output (modulo trivial timestamp metadata).

7. Technical Considerations

7.1 Bathymetric data and mask polygon

The bathymetric grid is the analytical bottleneck of this package. Its values are plausible at basin scale but they are not measurements; any quantitative claim about a specific berth, vessel, or approach depends on replacing the grid with measured data before operational use. Replacement requires placing a EPSG:32632 GeoTIFF in the raster source folder and re-running the pipeline from the spatial-analysis stage onwards; every downstream raster and depth-dependent vector layer regenerates against the new input. The harbour-basin mask polygon that constrains the synthetic grid (the main basin, the Darse Sud, the entrance fairway) is hand-defined and sets every pixel outside it to NoData. The mask confines all analytical outputs to the real water area and keeps the surrounding OpenStreetMap basemap visible behind transparent NoData pixels.

7.2 Coordinate handling

The pipeline ingests positions on the WGS 84 ellipsoid, performs every distance and area calculation in UTM 32N, and writes vector outputs in both systems so downstream consumers do not have to reproject.

Web GIS consumers (Cesium for Unreal, web map tile engines) read the WGS 84 GeoJSON directly; desktop GIS consumers and simulator engines that expect metric units read the UTM Shapefile. The two outputs describe the same features and were verified to agree to the precision the export format allows.

7.3 Cartographic limitations

The chart's symbology follows the colour convention of the International Hydrographic Organization's S-52 presentation library, with one important qualification: the colours are sRGB approximations rather than the formal CIE colour-space tokens that an S-52-certified Electronic Chart Display and Information System (ECDIS) renders. Treating the chart as anything more than a portfolio illustration would require both measured bathymetry (Section 7.1) and S-52-conformant colour rendering. The chart is an accurate map of an open-source data set, not a navigational publication.

7.4 Berth and route placeholders

Two scalar attributes in the berth specifications are placeholders: the berth_length column is fixed at 60.0 m for every berth, and the single safe-route polyline is a straight line between the harbour entrance and the deepest berth. The first requires tracing each berth's actual extent from current high-resolution imagery. The second requires running a least-cost-path solver against the cost-surface raster, which is already produced and ready for input.

8. Conclusion

The Port Hercule maritime GIS data package combines real OpenStreetMap features with a synthetic bathymetric grid and runs them through six analytical steps to produce nine vector layers, seven raster layers, a tabular berth specification and a complete cartographic chart, all in coordinate systems chosen for downstream consumption by simulator engines and web GIS tools. Every output passes an automated verification routine. The boundary between what the package supports and what it does not is set by one assumption: that bathymetric depth at any specific point can be replaced with a measured value. Section 7.1 describes the replacement procedure, which is the single step that promotes the work from portfolio demonstration to operational reference.

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